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B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

ME 19-205-0601 DYNAMICS OF MACHINERY (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Recall the concepts of free body diagrams, principles of statics and dynamics.
 CO2: Use graphical and analytic methods to do force analysis of planar mechanisms.
 CO3: Apply these concepts in different machine elements for the evaluation of forces and Moments.
 CO4: Analyze the dynamics of different mechanisms and machine elements and determine the various forces and torques.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

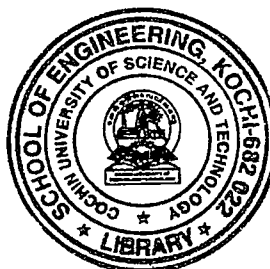
PO – Programme Outcome

PART A

(Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) What are the conditions for a body to be in equilibrium under the action of two forces, three forces and two forces and a torque?	3	3	L2	1	1,2
(b) What is meant by Piston effort and Crank effort?	3	3	L2	1	1,2
(c) The mass of flywheel of an engine is 6.5 tonnes and the radius of gyration is 1.8 m. It is found from the turning moment diagram that the fluctuation of energy is 56 kN-m. If the mean speed of the engine is 120 rpm, Find the maximum and minimum speeds.	3	3	L3	4	1,2
(d) What do you understand by Gyroscopic couple? Explain.	3	3	L2	4	1,2
(e) Explain Hammer blow. Find the maximum permissible angular speed of wheel if the balancing mass is 30 kg at a radius of 0.2 m and static wheel load is 15 kN.	3	3	L3	3	1,2
(f) A shaft rotating at 3000 rpm with an unbalanced mass of 15 kg at an eccentricity of 0.2 m need to be balanced by two masses M_1 and M_2 rotating in different planes. The plane of the 15 KG mass is at a distance of 200 mm to the right of the plane carries mass M_1 , and at a distance of 300 mm to the left of the plane carries mass M_2 . If both the balancing masses are at a radius of 0.35 m from the axis of shaft, find the magnitude of masses M_1 and M_2 .	3	3	L3	3	1,2
(g) Derive an expression for the velocity ratio of belt drive.	3	3	L3	4	1,2
(h) Define:	3	3	L2	4	1,2
(i) Sensitiveness of governor					
(ii) Stability of governor.					

(P.T.O.)



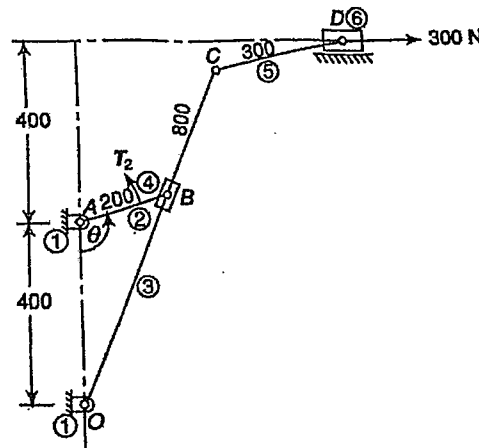
PART B

(4 × 12 = 48)

- II. The dimensions of a four-link mechanism are $AB = 500$ mm, $BC = 660$ mm, $CD = 560$ mm and $AD = 1000$ mm. the link AB has an angular velocity of 10.5 rad/sec counter clockwise and an angular retardation of 26 rad/sec² at the instant when it makes an angle 60° with AD, the fixed link. The masse of links BC and CD is 4.2 kg/m length. The link AB has a mass of 3.54 kg, the centre of which lies at 200 mm from A and a moment of inertia of 88500 kg mm². Neglecting the gravity and inertia effects, determine the instantaneous value of the drive torque required to be applied on AB to overcome the inertia forces.

OR

- III. For the static equilibrium of the quick return mechanism shown in figure, find the maximum input torque T_2 required for a force of 300 N on the slider D. Angle $\theta = 105^\circ$ Neglect friction.



All dimensions in mm

- IV. The turbine rotor of a ship has a mass of 3500 kg. it has a radius of gyration of 0.45 m and a speed of 3000 rpm clockwise when looking from stern. Determine the gyroscopic couple and its effect upon the ship.
- When the ship is steering to the left on a curve of 100 m radius at a speed of 36 km/h.
 - When the ship is pitching in a simple harmonic motion, the bow falling with its maximum velocity. The period of pitching is 40 seconds and the total angular displacement between the two extreme positions of pitching is 12 degrees.

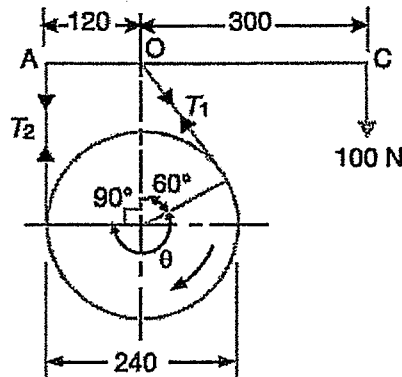
OR

(Continued)

BTS-VI(R/S)-05-24-2436

Marks	BL	CO	PO
5	L3	4	1,2,3

- V. (a) The simple band brake, as shown in Figure is applied to a shaft carrying a flywheel of mass 400 kg. The radius of gyration of the flywheel is 450 mm and runs at 300 rpm. If the coefficient of friction is 0.2 and the brake drum diameter is 240 mm, Find:
- The torque applied due to a hand load of 100 N
 - The number of turns of the wheel before it is brought to rest
 - The time required to bring it to rest, from the moment of the application of the brake.



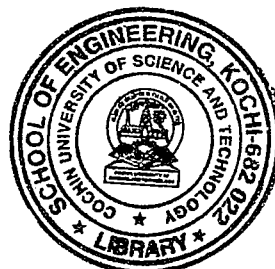
All dimensions in mm.

- (b) The turning moment diagram for a petrol engine is drawn to the following scale: Turning moment, 1 mm = 5 N-m; crank angle, 1 mm = 1°. The turning moment diagram repeats itself at every half revolution of the engine and the areas above and below the mean turning moment line taken in order are 295, 685, 40, 340, 960, 270 mm². The rotating parts are equivalent to a mass of 36 kg at a radius of gyration of 150 mm.
- Draw the turning moment diagram
 - Determine the coefficient of fluctuation of speed when the engine runs at 1800 rpm.
- VI. The three cranks of a three-cylinder locomotive are all on the same axle and are set at 120°. The pitch of the cylinder is 1m and the stroke of each piston is 0.6 m. The reciprocating masses are 300 kg for inside cylinder and 260 kg for each outside cylinder and the planes of rotation of the balance masses are 0.8 m from the inside crank. If 40% of the reciprocating parts are to be balanced, find:
- The magnitude and position of the balancing masses required at a radius of 0.6 m.
 - The hammer blow per wheel when the axle makes 6 rps.
- OR
- VII. A, B, C and D are four masses carried by a rotating shaft at radii 100 mm, 125 mm, 200 mm, and 150 mm respectively. The planes in which the masses revolve are 600 mm apart and the mass of B, C, and D are 10 kg, 5 kg, and 4 kg respectively. Find the required mass A and the relative angular settings of the four masses so that the shaft shall be in complete balance.

7	L4	4	1,2,3
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12	L4	3	1,2,3
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12	L4	3	1,2,3
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(P.T.O.)

BTS-VI(R/S)-05-24-2436

		Marks	BL	CO	PO
VIII.	(a) Find the power transmitted by a belt running over a pulley of 600 mm diameter at 200 rpm. The coefficient of friction between the belt and the pulley is 0.25, angle of lap is 160° and maximum tension in the belt is 2500 N.	5	L3	4	1,2,3
	(b) A rope drive transmits 600 kW from a pulley of effective diameter 4 m, which runs at a speed of 90 rpm. The angle of lap is 160° , angle of groove is 45° , the coefficient of friction is 0.28, the mass of rope is 1.5 kg/m and the allowable tension in each rope is 2400 N. Find the number of ropes required.	7	L3	4	1,2,3
OR					
IX.	(a) Calculate the vertical height of a Watt governor when it rotates at 60 rpm. Also find the change in vertical height when its speed increases to 61 rpm	5	L3	4	1,2,3
	(b) A porter governor has equal arms each 250 mm long and pivoted on the axis of rotation. Each ball has a mass of 5 kg and the mass of the central load on the sleeve is 25 kg. The radius of rotation of the ball is 150 mm when the governor begins to lift and 200 mm when the governor is at maximum speed. Find the minimum and maximum speeds and range of speed of the governor.	7	L3	4	1,2,3

Blooms's Taxonomy Level's

L2 – 10%, L3 – 34.2%, L4 – 55.8%.

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B.Tech. Degree VI Semester Special Supplementary Examination November 2022

ME 19-205-0601 DYNAMICS OF MACHINERY
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Recall the concepts of free body diagrams, principles of statics and dynamics.
 CO2: Use graphical and analytic methods to do force analysis of planar mechanisms.
 CO3: Apply these concepts in different machine elements for the evaluation of forces and moments.
 CO4: Analyze the dynamics of different mechanisms and machine elements and determine the various forces and torques.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer **ALL** questions)

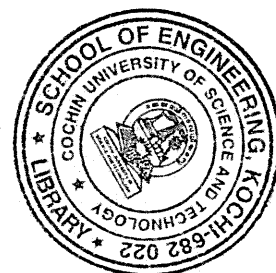
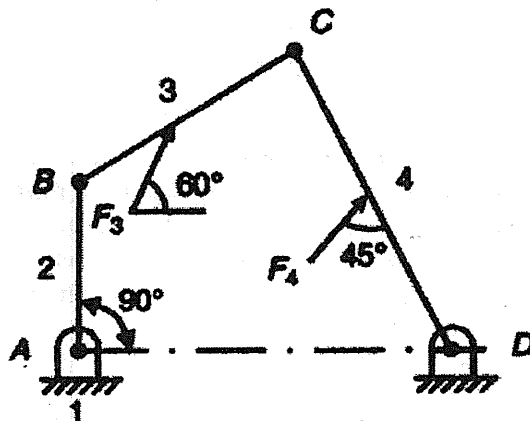
(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) State and explain D'Alembert's principle.	3	L2	1	1
(b) What is meant by piston and crank effort?	3	L1	2	1,2
(c) Define coefficient of fluctuation of energy and coefficient fluctuation of speed.	3	L1	3	1,2,3
(d) Explain the effect of gyroscopic couple on a naval ship.	3	L2	3	1,2,3
(e) What do you mean by primary and secondary unbalance in reciprocating engines?	3	L1	3	1,2,3
(f) Define hammer blow, tractive effort and swaying couple.	3	L2	3	1,2,3
(g) Explain the phenomena of 'slip' and 'creep' in a belt drive.	3	L2	3	1,2,3
(h) Explain the meaning of sensitiveness, hunting and stability of a governor.	3	L2	3	1,2,3

PART B

(4 × 12 = 48)

- II. (a) A four bar mechanism as shown in figure is subjected to two forces, $F_3 = 1000$ N at 60° from horizontal and $F_4 = 2000$ N at 45° from link 4. The dimensions of links are as under:
 $AB = 300$ mm, $BC = 400$ mm, $CD = 450$ mm and $AD = 600$ mm
 Perform static force analysis and determine resisting torque on link 2.



- (b) Define the terms inertia force and inertia couple. How is their magnitude and direction found?

OR

(P.T.O.)

		Marks	BL	CO	PO
III.	(a) A vertical cylinder petrol engine has a bore of 100 mm and stroke of 120 mm. The length of connecting rod between centres is 250 mm. The mass of the piston is 1.1 kg. The speed of engine is 1500 rpm. In the expansion stroke with a crank at 30° from TDC, the gas pressure is 700 kN/m^2 . Determine: (i) net force on the piston (ii) force on the connecting rod (iii) thrust on the cylinder wall (iv) crank effort (v) speed above which the gudgeon pin force would reverse in direction.	9	L4	4	1,2,3
	(b) What is dynamically equivalent system?	3	L1	1	1
IV.	(a) Write the procedure for determining the turning moment diagram.	5	L1	3	1,2,3
	(b) A double-acting steam engine develops 56 kW of power at 210 rpm. The maximum and minimum speeds do not vary more than 1% of the mean speed and the excess energy is 30% of the indicated work per stroke. Determine the mass of the flywheel if the radius of gyration of the flywheel is 500 mm.	7	L4	3	1,2,3
OR					
V.	(a) The rotor of a marine turbine has a moment of inertia of 750 kg.m^2 and rotates at 3000 rpm clockwise when viewed from aft. If the ship pitches with angular simple harmonic motion having a periodic time of 16 seconds and an amplitude of 0.1 radian, find: (i) the maximum angular velocity of the rotor axis (ii) the maximum value of the gyroscopic couple (iii) the gyroscopic effect as the bow dips	7	L3	3	1,2,3
	(b) How a four-wheeled vehicle is affected by gyroscopic couple?	5	L1	3	1,2,3
VI.	Four masses 150, 250, 200 and 300 kg are rotating in the same plane at radii of 0.25, 0.2, 0.3 and 0.35 m respectively. Their angular location is 40° , 120° and 250° from mass 150 kg, respectively, measured in counter-clockwise direction. Find the position and magnitude of the balance mass required, if its radius of rotation is 0.25 m.	12	L3	4	1,2,3
OR					
VII.	A single cylinder reciprocating engine has speed 240 rpm, stroke 300 mm, mass of reciprocating parts 5 kg and mass of revolving parts at 150 mm radius is 3.7 kg. If two-third of the reciprocating parts and all the revolving parts are to be balanced, find: (i) the balancing mass required at a radius of 400 mm (ii) the residual unbalanced force when the crank has rotated 60° from top dead centre.	12	L3	4	1,2,3
VIII.	Two parallel shafts that are 3.5 m apart are connected by two pulleys of 1-m and 400-mm diameters, the larger pulley being the driver runs at 220 rpm. The belt weighs 1.2 kg per metre length. The maximum tension in the belt is not to exceed 1.8 kN. The coefficient of friction is 0.28. Owing to slip on one of the pulleys, the velocity of the driven shaft is 520 rpm only. Determine: (i) the torque on each shaft (ii) the power transmitted (iii) the power lost in friction (iv) the efficiency of the drive.	12	L3	4	1,2,3

OR

(Continued)

BTS-VI(S.S)-11-22-1105

		Marks	BL	CO	PO
IX.	(a) What is the main function of a governor? How does it differ from that of a flywheel?	4	L2	4	1,2,3
	(b) Each arm of a Porter governor is 200 mm long and is pivoted on the axis of the governor. The radii of rotation of the balls at the minimum and the maximum speeds are 120 mm and 160 mm respectively. The mass of the sleeve is 24 kg and each ball is 4 kg. Find the range of speed of the governor. Also determine the range of speed if the friction at the sleeve is 18 N.	8	L3	4	1,2,3

Bloom's Taxonomy Levels

L1 = 20.8%, L2 = 15.8%, L3 = 42.5%, L4 = 20.8%.

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A

B.Tech. Degree VI Semester Regular Examination April 2022

ME 19-205-0601 DYNAMICS OF MACHINERY (2019 Scheme)

Time: 3 Hours

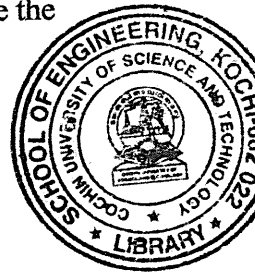
Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Recall the concepts of free body diagrams, principles of statics and dynamics
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Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome



PART A (Answer ALL questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	Explain the conditions for a body to be in equilibrium under the action of two forces, three forces and two force and torque.	3	L2	1	1
	(b)	The inertia of the connecting rod can be replaced by two masses concentrated at two points and connected rigidly together. How to determine the two masses so that it is dynamically equivalent to the connecting rod? Justify:	3	L1	1	2
	(c)	Explain the turning moment diagram of a multicylinder engine.	3	L2	4	1
	(d)	Explain the concept of gyroscopic couple? Derive the formula for its magnitude.	3	L2	4	2
	(e)	Define the terms 'static balancing' and 'dynamic balancing'. State the necessary conditions to achieve them.	3	L1	3	1
	(f)	Derive the following expressions, for an uncoupled two cylinder locomotive engine:	3	L1	4	2
		(i) Variation is tractive force				
		(ii) Swaying couple				
		(iii) Hammer blow				
	(g)	Show that the speed at which a belt or rope should be run to transmit maximum power is that at which the maximum allowable tension is three times the centrifugal tension in the belt or rope at that speed.	3	L1	4	3
	(h)	Explain stability of a governor? Sketch the controlling force versus radius diagrams for a stable, unstable and isochronous governor.	3	L2	4	2

PART B

(4 × 12 = 48)

II.	Determine the required input torque on the crank of a Slider-crank mechanism for static equilibrium when the applied piston load is 1500 N. The length of crank and connecting rod are 40 mm and 100 mm respectively and the crank is turned 45° from inner dead centre.	12	L3	2	2
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OR

(P.T.O.)

BTS-VI(R)-04-22-0282

- III. A horizontal steam engine running at 120 r.p.m. has a bore of 250 mm and a stroke of 400 mm. The connecting rod is 0.6 m and mass of the reciprocating parts is 60 kg. When the crank has turned through an angle of 45° from the inner dead Centre, the steam pressure on the cover end side is 550 kN/m^2 and that on the crank end side is 70 kN/m^2 . Considering the diameter of the piston rod equal to 50 mm, determine:
- turning moment on the crank shaft
 - thrust on the bearings
 - acceleration of the flywheel, if the power of the engine is 20 kW, mass of the flywheel 60 kg and radius of gyration 0.6 m.
- 12 L3 1 2
- IV. The turning moment diagram of a four stroke engine may be assumed for the sake of simplicity to be represented by four triangles in each stroke. The areas of these triangles are as follows:
- Suction stroke = $5 \times 10^{-5} \text{ m}^2$; Compression stroke = $21 \times 10^{-5} \text{ m}^2$;
Expansion stroke = $85 \times 10^{-5} \text{ m}^2$; Exhaust stroke = $8 \times 10^{-5} \text{ m}^2$
- All the areas excepting expansion stroke are negative. Each m^2 of area represents 14 MN-m of work. Assuming the resisting torque to be constant, determine the moment of inertia of the flywheel to keep the speed between 98 r.p.m. and 102 r.p.m. Also find the mass of a rim-type flywheel, given that density of flywheel material is 8150 kg/m^3 ; the allowable tensile stress of the flywheel material is 7.5 MPa.
- 12 L3 4 2
- OR
- V. A ship propelled by a turbine rotor which has a mass of 5 tonnes and a speed of 2100 r.p.m. The rotor has a radius of gyration of 0.5 m and rotates in a clockwise direction when viewed from the stern. Find the gyroscopic effects in the following conditions:
- The ship sails at a speed of 30 km/h and steers to the left in a curve having 60 m radius.
 - The ship pitches 6 degree above and 6 degree below the horizontal position. The bow is descending with its maximum velocity. The motion due to pitching is simple harmonic and the periodic time is 20 seconds.
 - The ship rolls and at a certain instant it has an angular velocity of 0.03 rad/s clockwise when viewed from stern.
- Determine the maximum angular acceleration during pitching. Explain how the direction of motion due to gyroscopic effect is determined in each case.
- 12 L3 4 2

(Continued)

- VI. A shaft carries four masses in parallel planes A, B, C and D in this order along its length. The masses at B and C are 18 kg and 12.5 kg respectively, and each has an eccentricity of 60 mm. The masses at A and D have an eccentricity of 80 mm. The angle between the masses at B and C is 100° and that between the masses at B and A is 190° , both being measured in the same direction. The axial distance between the planes A and B is 100 mm and that between B and C is 200 mm. If the shaft is in complete dynamic balance, determine:
- the magnitude of the masses at A and D
 - the distance between planes A and D
 - the angular position of the mass at D.
- OR**
- VII. The following data apply to an outside cylinder uncoupled locomotive:
- Mass of rotating parts per cylinder = 360 kg;
 Mass of reciprocating parts per cylinder = 300 kg ;
 Angle between cranks = 90° ;
 Crank radius = 0.3 m ;
 Cylinder centres = 1.75 m ;
 Radius of balance masses = 0.75 m ;
 Wheel centres = 1.45 m.
- If whole of the rotating and two-thirds of reciprocating parts are to be balanced in planes of the driving wheels, find:
- magnitude and angular positions of balance masses
 - speed in kilometres per hour at which the wheel will lift off the rails when the load on each driving wheel is 30 kN and the diameter of tread of driving wheels is 1.8 m
 - swaying couple at speed arrived at in above case.
- VIII. The following data refer to an open belt drive:
- Diameter of larger pulley = 400 mm ;
 Diameter of smaller pulley = 250 mm ;
 Distance between two pulleys = 2 m ;
 Coefficient of friction between smaller pulley surface and belt = 0.4 ;
 Maximum tension when the belt is on the point of slipping = 1200 N.
- Find the power transmitted at speed of 10 m/s. It is desired to increase the power. Which of the following two methods you will choose?
- Increasing the initial tension in the belt by 10 per cent
 - Increasing the coefficient of friction between the smaller pulley surface and belt by 10 per cent by the application of suitable dressing on the belt.
- Also find the percentage increase in power possible in each case.
- OR**
- IX. A Porter governor has equal arms each 250 mm long and pivoted on the axis of rotation. Each ball has a mass of 5 kg and the mass of the central load on the sleeve is 25 kg. The radius of rotation of the ball is 150 mm when the governor begins to lift and 200 mm when the governor is at maximum speed. Find the range of speed, sleeve lift, governor effort and power of the governor in the following cases:
- when the friction at the sleeve is neglected
 - when the friction at the sleeve is equivalent to 10 N.

Bloom's Taxonomy Levels

L1-25 , L2-25%, L3-43.75%, L4-6.25%, L5-0%

B.Tech. Degree VI Semester Special Supplementary Examination January 2019

ME 15-1601 DYNAMICS OF MACHINERY (2015 Scheme)

Time : 3 Hours

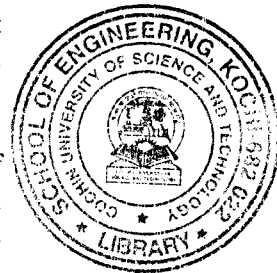
Maximum Marks : 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

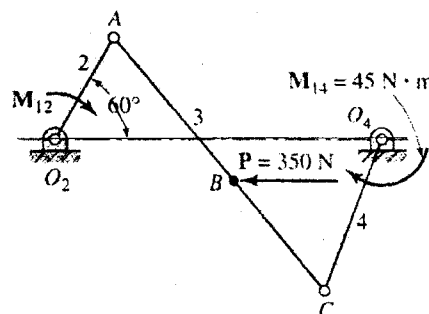
- I. (a) A horizontal engine has crank length 60 mm, connecting rod 240 mm and piston diameter 100 mm. When the crank has turned 30° from the inner dead centre, what will be the turning moment on the crank shaft due to a gas pressure of 580 kN/m².
- (b) A rod of mass 10 kg and length 1m, has a radius of gyration of 300 mm about its centre of gravity, G. Point G is at a distance of 400 mm from one end. If this rod is to be replaced by a dynamically equivalent system consisting of a 6 kg mass and a 4 kg mass, find the distances to these masses from G.
- (c) The rim of a flywheel has a rectangular cross section, width 200 mm and thickness 40 mm. It has a mean diameter of 1.4 m. If the tensile stress, the rim material can withstand is 18 N/mm², what is the maximum speed the flywheel can safely run? The rim material has a density of 7000 kg/m³.
- (d) Derive the expression for the ratio of tight side to slack side tensions for a band and block brake having n blocks.
- (e) A wheel of mass 6 kg and radius of gyration 250 mm, spinning at 300 rpm, is stopped using a block brake in 30 seconds. What will be the heat generated in the braking process?
- (f) There are three masses A, B and C revolving in a plane perpendicular to a shaft. They have masses 4 kg, 5 kg and 5 kg respectively and are attached to the shaft by arms which have lengths 300 mm, 400 mm and 300 mm respectively. Find the angle between the arms for complete balance.
- (g) Distinguish between static balancing and dynamic balancing.
- (h) What is the maximum torque that can be transmitted by the smaller pulley of 300 mm radius, if the angle of wrap of the belt around it is 160°, the initial tension in the belt is 1000 N, and the coefficient of friction between the belt and the pulley is 0.25.
- (i) In a chain drive, for constant angular velocity for the driving sprocket, will the angular velocity of driven sprocket remain constant? Why?
- (j) With a neat sketch explain how an inertia governor functions? What are its advantages over centrifugal governor?



PART B

(4 × 10 = 40)

- II. Determine the torque M_{12} to be applied to the link 2 of the four link mechanism shown in figure to maintain static equilibrium at the given position. Lengths $O_2A=200$ mm, $AB=400$ mm, $AC=700$ mm, $O_2O_4=700$ mm, $O_4C=350$ mm.



OR

(P.T.O.)

- III. The connecting rod of a vertical high speed engine has a mass of 2.5 kg and is 30 cm long between centres. Its centre of gravity is 10 cm from the big end bearing. When suspended in a vertical plane and allowed to swing about the axis of the small end bearing, it makes 50 complete oscillations in 53 seconds. The stroke of the piston is 12.5 cm. The reciprocating parts have a mass of 1 kg. Determine the torque exerted on the crank shaft due to inertia of the moving parts when crank makes an angle of 140° with the top dead centre and the speed of rotation is 1600 rpm.

- IV. A shaft fitted with a flywheel rotates at 250 rpm and drives a machine. The torque of the machine varies in a cyclic manner over a period of 3 revolutions. The torque rises from 750 Nm to 3000 Nm uniformly during $\frac{1}{2}$ revolutions and remains constant for the following revolution. It then falls uniformly to 750 Nm during the next $\frac{1}{2}$ revolution and remains constant for one revolution, this being repeated thereafter. Determine the power required to drive the machine and percentage fluctuation of the speed, if the driving torque applied to shaft is constant and the mass of the flywheel is 500 kg and radius of gyration 60 cm.

OR

- V. A rail car having a total mass of 4000 kg is negotiating a curve of 300 m radius, where the track is banked at an angle of 10° . The car has two axles each of which together with its wheels and gearing has a total moment of inertia of 30 kg m^2 . The centre distance between the two wheels on an axle is 1.5 m and each wheel has 46.5 cm radius. Each axle is driven by a motor, the speed ratio between the two being 1:3. Each motor with its gear has a moment of inertia of 15 kg m^2 and runs in the direction opposite to that of the axle. The centre of gravity of the car is 105 cm above the rails when on level track. Determine the limiting speed for the car so that all the wheels maintain contact with the rail.

- VI. The intermediate cranks of a four cylinder symmetrical engine, which is in complete primary balance, are at 90° to each other and each has a reciprocating mass of 300 kg. The centre distance between the intermediate cranks is 700 mm and that between the extreme cranks is 1800 mm. Lengths of the cranks and connecting rods are 250 mm and 900 mm respectively. Calculate the masses fixed to the extreme cranks with their relative angular positions. Also find the secondary forces and couple if the engine runs at 600 rpm.

OR

- VII. Three cylinders of an air compressor have their lines of stroke 120° to one another and their connecting rods are coupled to a single crank. The stroke is 10 cm and the length of each connecting rod is 15 cm. Mass of the reciprocating parts per cylinder is 1.5 kg. Determine the primary and secondary unbalanced forces of the engine running at 3000 rpm.
- VIII. Power is transmitted by an open belt drive from a pulley 30 cm in diameter running at 200 rpm to a pulley 50 cm in diameter. Angle of lap on the small pulley is 160° . The belt is on the point of slipping when 2.6 kW is being transmitted. $\mu = 0.25$. It is desired to increase the power transmitted. State which of the following two methods suggested would be more effective: (i) Initial tension in the belt is increased by 10% (ii) Suitable dressing is given to the friction surface of the belt to increase the coefficient of friction by 10%.

OR

- IX. The mass of each ball of a Proell governor is 7.5 kg and the load on the sleeve is 80 kg. Each arm is 300 mm long. The upper arms are pivoted on the axis of rotation whereas the lower arms are pivoted to links of 40 mm from the axis of rotation. The extensions of the lower arms to which the balls are attached are 100 mm long and are parallel to the governor axis at the minimum radius. Determine the equilibrium speeds corresponding to extreme radii of 180 mm and 240 mm.

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B.Tech. Degree VI Semester Supplementary Examination April 2022

ME 15-1601 DYNAMICS OF MACHINERY

(2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(Answer *ALL* questions)

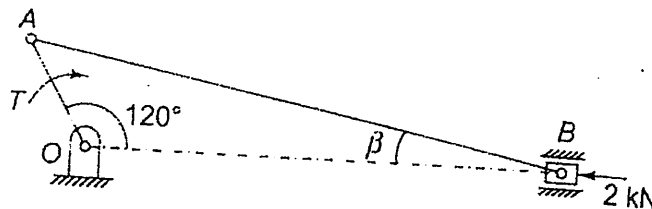
(10 × 2 = 20)

- I. (a) A link in a mechanism is connected to two other links by pin joints at its ends. Describe the effect of pin-friction on the force through the link.
- (b) Why a correction couple is used when a connecting rod is replaced by two point-masses?
- (c) Derive the gyroscopic couple and show its effect on an aircraft taking a turn.
- (d) Explain self-locking in a block brake using a neat sketch.
- (e) Derive an expression for braking torque in a band and block brake.
- (f) What is meant by dynamic balancing? What are the conditions to be met for dynamic balance?
- (g) Derive the expression for swaying couple. What is its effect on the locomotive?
- (h) Explain the effect of initial tension on the maximum torque transmitted by a belt drive.
- (i) Describe the different types of chains used in engineering.
- (j) Explain hunting and isochronism in governors.

PART B

(4 × 10 = 40)

- II. In the slider-crank mechanism shown in the figure, the link OA = 100 mm and AB = 450 mm. A force F = 2 kN acts on the slider at B. Determine the torque T required on the link OA for the static equilibrium of the mechanism.



OR

- III. In a vertical double-acting steam engine, the connecting rod is 4.5 times the crank. The weight of the reciprocating parts is 120 kg and the stroke of the piston is 440 mm. The engine runs at 250 rpm. If the net load on the piston due to steam pressure is 25 kN when the crank has turned through an angle of 120° from the top dead centre, determine (i) thrust in the connecting rod (ii) pressure on the slide bars (iii) tangential force on the crank pin (iv) thrust on the bearings (v) turning moment on the crank shaft.

(P.T.O.)

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- IV. The turning moment diagram for a petrol engine is drawn to a vertical scale of 1mm to 6 Nm and a horizontal scale of 1mm to 1° . The turning moment repeats itself after every half revolution of the engine. The areas above and below the mean torque line are + 305, - 710, +50, -350, +980 and -275 mm². The rotating parts amount to a mass of 40kg at a radius of gyration of 140 mm. Calculate the co-efficient of fluctuation of speed if the speed of the engine is 1500 rpm.

OR

- V. A motor cycle rider is moving along a curve of radius 35m at a speed of 72 km/h. The combined mass of the motor cycle and rider is 220 kg with the centre of mass 620 mm above the ground level when the motor cycle is upright. Each wheel of the motor cycle is 600 mm diameter and has a moment of inertia of 1.1 kg m². The moment of inertia of the rotating parts of the engine is 0.18 kg m². The engine rotates at 4.5 times the speed of road wheels in the same sense. Find the angle of heel.

- VI. A rotating shaft carries four masses A, B, C and D that are completely balanced. Masses C and D make angles of 90° and 195° respectively with that of mass B. The rotating masses have the following properties: masses $m_b = 25$ kg, $m_c = 40$ kg, $m_d = 35$ kg and radii $r_a = 150$ mm, $r_b = 200$ mm, $r_c = 100$ mm and $r_d = 180$ mm. Planes B and C are 250 mm apart. Determine (i) the mass A and its angular position with that of mass B (ii) positions of all planes relative to plane of mass A.

OR

- VII. The length of each connecting rod of a 60° V-engine is 220 mm and the stroke is 100 mm. The mass of the reciprocating parts is 1.2 kg per cylinder and the crank speed is 2400 rpm. Find the values of the primary and secondary forces.

- VIII. An open-belt drive is required to transmit 10 kW of power from a motor running at 600 rpm. Diameter of the driving pulley is 250 mm. The speed of driven pulley is 220 rpm. The belt is 12 mm thick and has a mass density of 0.001 g/mm³. Safe stress in the belt is not to exceed 2.5 N/mm². The two shafts are 1.25 m apart. The co-efficient of friction is 0.25. Determine the width required for the belt.

OR

- IX. Each arm of a Porter governor is 250 mm long. The upper and lower arms are pivoted to links of 40 mm and 50 mm respectively from the axis of rotation. Each ball has a mass of 5 kg and the sleeve mass is 50 kg. The force of friction on the sleeve of the mechanism is 40 N. Determine the range of speed of the governor for extreme radii of rotation of 125 mm and 150 mm.
